Geometric time-dependent density functional theory for lattice systems

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Recap of abstract theory

Setup:

- (Finite-dimensional) Hilbert space $\mathfrak{H}$, e.g., $\mathfrak{H} = \mathbb{C}^d$
- Hamiltonian, i.e. Hermitian matrix, $H(t)$ on $\mathfrak{H}$
- Family of (commuting) observables $\{\mathcal{O}_n\}_{n=1}^M$ and desired expectation values $\{o_n(t)\}_{n=1}^M$

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Then, **Variational (VP)** and **Geometric (GP)** principles are

VP Find functions $v_n(t)$ such that

$$i\partial_t \Psi(t) = \left(H(t) + \sum_{n=1}^{M} v_n(t)\mathcal{O}_n \right) \Psi(t)$$

has $\langle \mathcal{O}_n \rangle_{\Psi(t)} = o_n(t)$ for all t.

Possible if $K^{\Psi(t)}(t)$ invertible,

$$2K^{\Psi}_{mn}(t) = \langle [\mathcal{O}_n, [H(t), \mathcal{O}_m]] \rangle_{\Psi}$$

GP Find functions $w_n(t)$ such that

$$i\partial_t \Psi(t) = \left(H(t) + i\sum_{n=1}^{M} w_n(t)\mathcal{O}_n \right) \Psi(t)$$

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Possible if $S^{\Psi(t)}$ invertible,

$$S^{\Psi}_{mn} = \langle \mathcal{O}_n \mathcal{O}_m \rangle_{\Psi} = \langle \mathcal{O}_m \mathcal{O}_n \rangle_{\Psi}$$

Note: $K^{\Psi}(t)$ invertible $\implies$ S^{Ψ} invertible.

Lattice system

N spin-1/2 fermions on M sites $\{1, \ldots, M\}$. (Think $M = L^\nu$ a ν-dimensional box of side lengths L.)

- Hilbert space $\mathfrak{H} = \bigwedge^N (\mathbb{C} \times \{\uparrow, \downarrow\})^M$ of dimension $\binom{2M}{N} = \frac{(2M)!}{N!(2M-N)!}$
- Hamiltonian $H(t) = \underbrace{-\tau \sum_{m \sim m', \sigma} a^\dagger_{m\sigma} a_{m'\sigma}}_{T=} + \underbrace{\sum_{m,\sigma} v_{\mathrm{ext},m}(t) a^\dagger_{m\sigma} a_{m\sigma}}_{V_{\mathrm{ext}}(t)=} + \underbrace{\sum_{m,m'\sigma} U_{m'm} a^\dagger_{m\sigma} a_{m\sigma} a^\dagger_{m'\sigma'} a_{m'\sigma'}}_{V_{ee}=}$
- Observables $\mathcal{N}_m = a^\dagger_{m\uparrow} a_{m\uparrow} + a^\dagger_{m\downarrow} a_{m\downarrow}$.
- Constraints $\langle \mathcal{N}_m \rangle_{\Psi(t)} = \rho_m(t)$ for given $\{\rho_m(t)\}_{m=1}^M$

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- **VP** gives equation $i\partial_t \Psi^{\mathrm{VP}}(t) = (T + V_{\mathrm{ext}}(t) + V_{ee} + V(t))\Psi^{\mathrm{VP}}(t)$, with $V(t) = \sum_{m,\sigma} v_m(t) a^\dagger_{m\sigma} a_{m\sigma}$
- **GP** gives equation $i\partial_t \Psi^{\mathrm{GP}}(t) = (T + V_{\mathrm{ext}}(t) + V_{ee} + iW(t))\Psi^{\mathrm{GP}}(t)$, with $W(t) = \sum_{m,\sigma} w_m(t) a^\dagger_{m\sigma} a_{m\sigma}$

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Contents

- Formulate the Kohn–Sham scheme(s)
- Study invertibility of S^Ψ and K^Ψ —— When are they invertible for the initial condition Ψ_0?
- (Apply the Variational Principle to time-dependent *current* density functional theory (TDCDFT))

Standard Kohn–Sham

Definition (Hxc potential)

Define $V_0[\rho]$ and $V[\rho]$ via auxiliary systems

$$i\partial_t \tilde{\Psi} = (T + V_{ee} + V[\rho](t))\tilde{\Psi}(t), \qquad i\partial_t \tilde{\Phi} = (T + V_0[\rho](t))\tilde{\Phi}(t),$$

such that $\rho_{\tilde{\Psi}} = \rho_{\tilde{\Phi}} = \rho$. Then,

$$V_{\mathrm{Hxc}}[\rho] = V_0[\rho] - V[\rho].$$

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The **Kohn–Sham** scheme

$$i\partial_t \Phi(t) = (T + V_{\mathrm{ext}}(t) + V_{\mathrm{Hxc}}[\rho_\Phi](t))\Phi(t)$$

reproduces the density of the interacting system.

Proof.

By construction, $V_{\mathrm{ext}} + V_{\mathrm{Hxc}}[\rho_\Phi] = V_0[\rho_\Phi]$. Hence, $V[\rho_\Phi] = V_{\mathrm{ext}}$. By "Runge–Gross" the desired follows. □

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To determine V_{Hxc} more concretely solve for $V_0[\rho]$ (and $V[\rho]$) the equation(s) (van Leeuwen 1999)

$$\partial_t^2 \rho_m = \partial_t^2 \langle \mathcal{N}_m \rangle_{\tilde{\Psi}(t)} = 2\sum_n (K^{\tilde{\Psi}(t)})_{mn} (V_0[\rho](t))_n - \langle [T, [T, \mathcal{N}_m]] \rangle_{\tilde{\Psi}(t)}, \qquad \partial_t^2 \rho_m = \partial_t^2 \langle \mathcal{N}_m \rangle_{\tilde{\Phi}(t)} = \dots$$

Note: If interacting density is known, we can use that instead of ρ_Φ to calculate V_{Hxc}. Then clearly $V[\rho] = V_{\mathrm{ext}}$.

Geometric Kohn–Sham

Definition (Geometric Kohn–Sham potential)

Define $W_0[\rho]$ and $W[\rho]$ via auxiliary systems

$$i\partial_t \tilde{\Psi} = (T + V_{\mathrm{ext}}(t) + V_{ee} + iW[\rho](t))\tilde{\Psi}(t), \qquad i\partial_t \tilde{\Phi} = (T + V_{\mathrm{ext}}(t) + iW_0[\rho](t))\tilde{\Phi}(t),$$

such that $\rho_{\tilde{\Psi}} = \rho_{\tilde{\Phi}} = \rho$. Then,

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$$i\partial_t \Phi(t) = (T + V_{\mathrm{ext}}(t) + iW^{\mathrm{KS}}[\rho_\Phi](t))\Phi(t)$$

reproduces the density of the interacting system.

Proof.

By construction, $W^{\mathrm{KS}}[\rho_\Phi](t) = W_0[\rho_\Phi]$. Hence, $W[\rho_\Phi] = 0$. By "geometric Runge–Gross" the desired follows. □

Equivalent Neumann form:

$$i\partial_t \gamma(t) = \left[T + V_{\mathrm{ext}}(t) + i[W^{\mathrm{KS}}[\rho_\gamma](t), \gamma(t)], \gamma(t)\right].$$

Kohn–Sham with geometric correction

Let $V_{\mathrm{Hxc}}^{\mathrm{app}}[\rho]$ be any approximation of the Hxc potential. Then can consider the **Kohn–Sham scheme with geometric correction**

$$
\begin{aligned}
i\partial_t \tilde{\Phi}(t) &= (T + V_{\mathrm{ext}}(t) + V_{\mathrm{Hxc}}^{\mathrm{app}}[\rho_\Phi] + iW_0^{\mathrm{corr}}(t))\tilde{\Phi}(t) \\
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i\partial_t \Phi(t) &= (T + V_{\mathrm{ext}}(t) + V_{\mathrm{Hxc}}^{\mathrm{app}}[\rho_\Phi] + i(W_0^{\mathrm{corr}}(t) - W^{\mathrm{corr}}(t)))\Phi(t)
\end{aligned}
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The first two auxiliary equations define the functions W_0^{corr} and W^{corr} and the last the reproduces the interacting density as ρ_Φ.

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Problem

Determine $W^{\#}$ (in either geometric Kohn–Sham or Kohn–Sham with geometric correction)

- *as W_0 in terms of known true interacting density ρ, or*
- *as the self-consistent Kohn–Sham functional (in terms of W_0 and W above)*

Note: Both yield the same function $(W(t)_m)_{m=1}^M$. (As they must.)

In toy models where we can find the interacting density ρ exactly, it is easiest to use just the $W_0[\rho]$ functional.

How to determine W

Problem

Determine $W_0[\rho]$ given by $i\partial_t\tilde{\Phi} = (T + V_{\text{ext}}(t) + iW_0[\rho](t))\tilde{\Phi}(t)$ having density $\langle \mathcal{N}_m \rangle_{\tilde{\Phi}(t)} = \rho_m(t)$.

We compute

$$\partial_t \rho_m = \partial_t \langle \mathcal{N}_m \rangle_{\tilde{\Phi}(t)} = -\langle -i[T, \mathcal{N}_m] \rangle_{\tilde{\Phi}(t)} + 2\sum_{n=1}^{M} (S^{\tilde{\Phi}(t)})_{mn} (W_0[\rho](t))_n, \qquad \left(\text{recall } (S^{\tilde{\Phi}(t)})_{mn} = \langle \mathcal{O}_n \mathcal{O}_m \rangle_{\tilde{\Phi}(t)}\right).$$

The term $\langle -i[T, \mathcal{N}_m] \rangle_{\tilde{\Phi}(t)}$ should be though of as $(\operatorname{div} j_{\tilde{\Phi}(t)})_m$, the divergence of the current at site m.

Get "explicit" form

$$W_0[\rho](t) = \frac{1}{2} \left(S^{\tilde{\Phi}(t)}\right)^{-1} \left(\partial_t \rho + \operatorname{div} j_{\tilde{\Phi}(t)}\right)$$

Similar formula holds for auxiliary interacting system for $W[\rho]$.

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$\rightsquigarrow$ As long as we can invert $S^{\tilde{\Phi}(t)}$ (and $S^{\tilde{\Psi}(t)}$) we can solve the auxiliary system(s) and thus find the density of the interacting system from the Kohn–Sham scheme.

Problem

For what states ψ are S^{ψ} and K^{ψ} invertible?

Invertibility of S^ψ and K^ψ — I

Consider $\psi(t)$ either $\tilde{\Phi}(t)$ or $\tilde{\Psi}(t)$. By continuity in time: (ψ_0 denotes initial condition)

- S^{ψ_0} (resp. K^{ψ_0}) invertible $\implies$ $S^{\psi(t)}$ (resp. $K^{\psi(t)}(t)$) invertible for small times t.

Thus, study

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S^{ψ_0} is "generically" invertible:

Theorem

Suppose that $0 < \rho_m < 2$ for all $m = 1, \dots, M$, and that ψ_0 is an N-particle Slater determinant with $(\rho_{\psi_0})_m = \rho_m$.
Then, for any $\varepsilon > 0$, there exists a unitary matrix U_ε (on the one-particle space $(\mathbb{C} \times \{\uparrow, \downarrow\})^M$) with

$$\|1 - U_\varepsilon\| \leq \varepsilon, \qquad \rho_{\psi_0'} = \rho_{\psi_0}, \qquad S^{\psi_0'} \text{ invertible}, \qquad \text{where } \psi_0' = U_\varepsilon^{\otimes N} \psi_0.$$

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Thus, change bad initial condition $\psi_0 \rightsquigarrow$ better-behaved initial condition ψ_0' arbitrarily close to ψ_0.

Invertibility of S^ψ and K^ψ — II

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Invertibility of S^{ψ_0} (and K^{ψ_0}) are static problems and linked to other well-known problems in static DFT.

- S^{ψ_0} invertible $\implies$ *unique* v-representability (ψ_0 ground state of $H + V_1$ and of $H + V_2 \implies V_1 = V_2 + C$)
  ($\not\Longrightarrow$ $\rho = \rho_{\psi_0}$ is v-representability, i.e., that there exists ψ_0' ground state of some $H + V$ with $\rho_{\psi_0'} = \rho$)
- If ψ_0 non-degenerate ground state of H then

$$S^{\psi_0} \text{ invertible} \iff \text{density-density linear response function } \chi \text{ is invertible.}$$

- If ψ_0 non-degenerate ground state of H then ($\Longleftarrow$ always holds)

$$S^{\psi_0} \text{ invertible} \iff K^{\psi_0} \text{ invertible.}$$

All properties expected to hold "generically". Proving these for concrete states ψ_0 are difficult questions.

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Remark (H-dependence)

Any state ψ_0 is non-degenerate ground state of Hamiltonian $H = -|\psi_0\rangle\langle\psi_0|$.
All properties depend implicitly on the choice of reference Hamiltonian H:

- v-representability references particular perturbations of H (of the form $V = \sum v_m a^\dagger_{m\sigma} a_{m\sigma}$),
- Exactly same for linear response,
- Matrix $(K^{\psi_0})_{mn} = \frac{1}{2}\langle[\mathcal{N}_n, [H, \mathcal{N}_m]]\rangle_{\psi_0}$ depends on H, matrix $(S^{\psi_0})_{mn} = \langle\mathcal{N}_n\mathcal{N}_m\rangle_{\psi_0}$ does not.

Variational principle for time-dependent *current* density functional theory

Consider 1D chain and current-density and density operators

$$\mathcal{J}_m = i\tau \sum_\sigma \left(a^\dagger_{m\sigma} a_{(m+1)\sigma} - a^\dagger_{(m+1)\sigma} a_{m\sigma} \right), \qquad \mathcal{N}_m = a^\dagger_{m\uparrow} a_{m\uparrow} + a^\dagger_{m\downarrow} a_{m\downarrow}.$$

Apply VP with observables $\{\mathcal{O}_m\} = \{\mathcal{N}_m, \mathcal{J}_m\}_{m=1}^{M-1}$ (can remove $\mathcal{N}_M$ by gauge and $\mathcal{J}_M$ by continuity equation).

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No longer commuting observables $\rightsquigarrow$ consider matrix $A^\psi_{mn} = \frac{1}{2} \langle i[\mathcal{O}_n, \mathcal{O}_m] \rangle_\psi$. In our case

$$A^\psi = \begin{pmatrix} 0 & B^\psi \\ -(B^\psi)^T & C^\psi \end{pmatrix}, \qquad B^\psi_{mn} = \frac{1}{2} \langle i[\mathcal{J}_n, \mathcal{N}_m] \rangle_\psi, \qquad C^\psi_{mn} = \frac{1}{2} \langle i[\mathcal{J}_n, \mathcal{J}_m] \rangle_\psi$$

and

$$\det A^\psi = (\det B^\psi)^2 = \prod_{m=1}^{M-1} \left[\tau \sum_\sigma 2 \operatorname{Re} \left\langle a^\dagger_{(m+1)\sigma} a_{m\sigma} \right\rangle_\psi \right]^2.$$

If this is non-zero (generically true) then by the VP we can find $v_m(t)$ and $\alpha_m(t)$ such that

$$i\partial_t \Psi^{\mathrm{VP}}(t) = \left[T + V_{ee} + \sum_{m=1}^{M-1} \left(v_m(t) \mathcal{N}_m + \alpha_m(t) \mathcal{J}_m \right) \right] \Psi^{\mathrm{VP}}(t)$$

has $\langle \mathcal{N}_m \rangle_{\Psi^{\mathrm{VP}}(t)} = \rho_m(t)$ and $\langle \mathcal{J}_m \rangle_{\Psi^{\mathrm{VP}}(t)} = j_m(t)$ for given ρ and j.

Summary

Interacting Hamiltonian $H(t) = -\tau \sum_{m\sim m'} a^\dagger_{m\sigma} a_{m'\sigma} + \sum v_{\mathrm{ext},m}(t) a^\dagger_{m\sigma} a_{m\sigma} + \sum U_{m'm} a^\dagger_{m\sigma} a_{m\sigma} a^\dagger_{m'\sigma'} a_{m'\sigma'}$.

Geometric Principle: Given a density $(\rho_m(t))_{m=1}^M$ and initial condition $\psi_0 \in \mathfrak{H} = \bigwedge^N (\mathbb{C} \times \{\uparrow, \downarrow\})^M$ we can find $W(t) = \sum w_m(t) a^\dagger_{m\sigma} a_{m\sigma}$ such that

$$i\partial_t \psi(t) = (H(t) + iW(t))\,\psi(t), \qquad \psi(t=0) = \psi_0$$

has $\langle \mathcal{N}_m \rangle_{\psi(t)} = \rho_m(t)$. The vector $(w_m(t))_{m=1}^M$ is given by

$$(w_m(t))_{m=1}^M = \frac{1}{2}\left(S^{\psi(t)}\right)^{-1}\left(\partial_t \rho + \operatorname{div} j_{\psi(t)}\right), \qquad (S^{\psi(t)})_{mn} = \langle \mathcal{N}_m \mathcal{N}_n \rangle_{\psi(t)}, \qquad (\operatorname{div} j_{\psi(t)})_m = \langle -i[T, \mathcal{N}_m] \rangle_{\psi(t)}.$$

Geometric Kohn–Sham scheme: Geometric Kohn–Sham equation

$$i\partial_t \Phi(t) = (T + V_{\mathrm{ext}}(t) + i(W_0 - W))\Phi(t), \qquad \Phi(t=0) = \Phi_0$$

reproduces the density of interacting system, where W_0 and W are as in the GP above with density $\rho = \rho_\Phi$ and Hamiltonians without and with interaction V_{ee}.

If we already know the density of the interacting system, we use this instead and have $W = 0$.